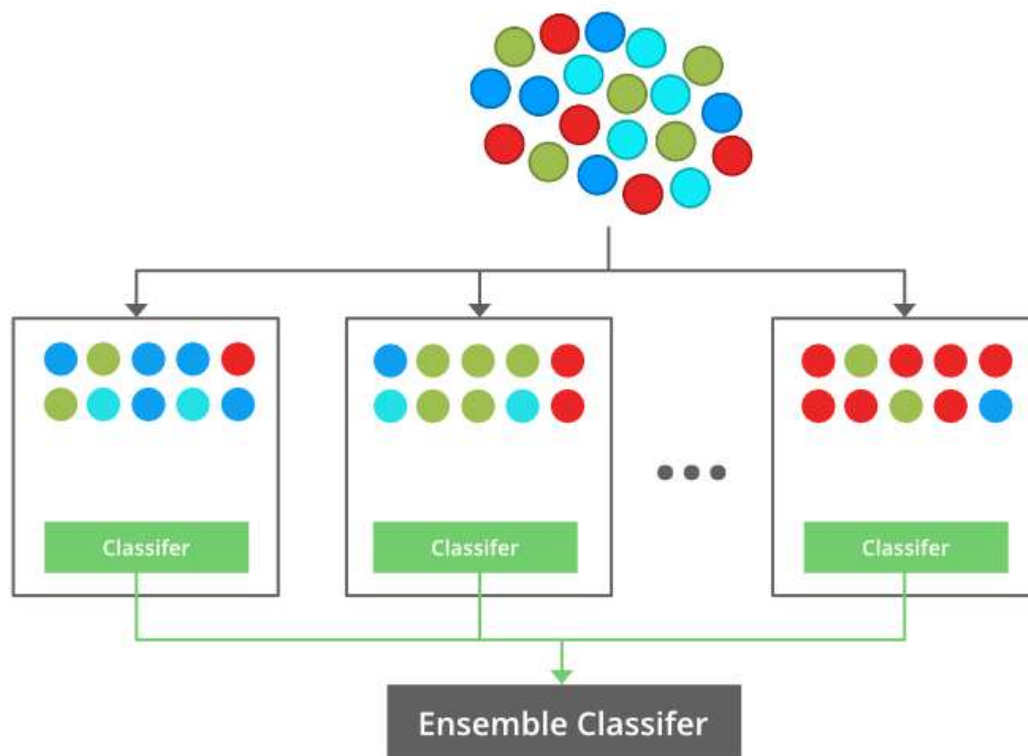


What is XGBoost?

- XGBoost stands for Extreme Gradient Boosting.
- It is a powerful machine learning algorithm.
- Used for classification and regression problems.
- Builds strong predictions using many small decision trees.

How XGBoost Works

- Step 1: Train the first decision tree.
- Step 2: Find prediction errors.
- Step 3: Build a new tree to correct mistakes.
- Step 4: Repeat the process many times.
- Step 5: Combine all trees for final prediction.



Original Data

Bootstrapping

Aggregating

Bagging

Advantages and Disadvantages

- Advantages:
 - - High performance and speed
 - - Handles missing data well
- Disadvantages:
 - - More complex than simple models
 - - Training can take longer on very large data

Simple Real-Life Example

- Imagine a cricket coach training players.
- After each match, the coach identifies mistakes.
- The next training session focuses on improving those mistakes.
- XGBoost improves predictions in the same way step by step.